Harsh Upadhyay

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Education

McMaster University | Honours Computer Science Co-op + Minor in Statistics — Sep 2022 – Apr 2027

- **Courses:** Data Struct & Algorithms, Operating Systems, Databases, Compilers, Machine Learning, NLP, Cryptography

Research

McMaster University | Research Assistant (Supervisor: Prof. Cansu Ekmekcioglu) — Sep 2026 – Present

- Built **Palimpsest** (**Node.js, Express, Turso**) to trace changes across **740+ Wayback captures** via **CDX** digest collapsing.
- Engineered a **DOM block-diff engine** (**git-style sequence alignment**, **URL canonicalization**) with **sandboxed iframes**.
- Cut timelapse load from **740 to 14 requests** and cold diffs from **36.5s to 3.5s** via **IndexedDB** and **dual-queue rate limiting**.

OCAD University | Research Assistant (Supervisor: Prof. Sara Diamond) — April 2026 – December 2026

- Benchmarked **BPE** and **unigram LM** tokenizers against **morphology-aware** segmentation for polysynthetic languages.
- Implemented **SingSong** (Google, 2023): **T5** and decoder-only **Transformers** over **w2v-BERT** and **SoundStream RVQ** tokens.
- Lifted key detection from **55% to 93.2%** on **OpenEWLD** via a 24-way classifier feeding **semi-Markov Viterbi** chord decoding.
- Replaced **Whisper** with fused **MMS-1b** Plains Cree **ASR** adapters over **Demucs**-isolated vocals to stop English hallucination.
- Engineered a real-time harmonizer: **YIN/RMVPE** tracking, **10 accompaniment modes**, and **RVC** voice conversion in a sidecar.

Experience

Government of Ontario - MPBSDP | IT Systems Assistant — May 2026 – August 2026

- Drove **OAC DV** rollout to **200+ users** via technical workshops, **workbook development**, and **AI feature evaluation**.
- Optimized **OAC subject areas**, **Logical SQL**, and **workbooks** to speed ministry-wide **self-service analytics**.
- Automated **Detailed Schedules of Payments** validation across ministry extracts with **SQL** and **Python** for **Public Accounts**.
- Built an interactive **APEX Procurement and Contract Tracking** dashboard turning raw data into **actionable insights**.

Government of Ontario - MPBSDP | Business Analyst Co-op — May 2025 – December 2025

- Raised Match and Merge automated throughput from **30% to 70%** by leading **Agile** refinement and writing **Jira** user stories.
- Validated **REST APIs** and third-party integrations end to end in **Postman**, catching defects and enforcing business rules.
- Assessed **migration feasibility** via **Python-SQL**, matching legacy postal records to external data and **GeoJSON boundaries**.

Projects

Recur — *React, TypeScript, Node.js, Express, Ollama*

- Built a **program synthesis compiler** turning repeated **LLM** queries into **typed programs** answered in $<$**1ms**.
- Designed a **best-first search** over a **dataflow IR** (13 readers, 41 transforms) ranking **12,000+** candidates by **simplicity**.
- Matched requests with **256-dim hashed n-gram vectors** and hybrid scoring, gating reuse via a **neural acceptance model**.
- Verified programs by **leave-one-out** generalization and **64-input** stress tests, declining back to the **LLM** when unprovable.

Arena — *Next.js, TypeScript, Tailwind CSS, Go, WebSockets, OpenAI, Anthropic, Groq*

- Engineered a **Go** backend with a **unified LLM client** (OpenAI, Anthropic, Groq) running live **Wordle, Connect 4, Codenames**.
- Isolated each match in a **goroutine**; concurrent Wordle agents share **immutable state**, eliminating the need for **locks**.
- Built a resilient **LLM output parser** stripping reasoning-model **think blocks**, with **3x retries** under stricter prompts.
- Streamed matches via a **WebSocket hub** with **per-match rooms** to a **Next.js 15** dashboard with a live **ELO leaderboard**.

ActiveTrack — *Python, Pandas, NumPy, Scikit-Learn, Matplotlib*

- Built an **ML classifier** reaching **98.9%** accuracy on 5 barbell exercises from **IMU sensor** data (**accelerometer + gyroscope**).
- Engineered **25+ features** using **Chauvenet outlier removal**, **low-pass filtering**, **PCA**, **FFT**, and **K-means** clustering.
- Selected models via **forward selection** and **grid search** over **Random Forest**, **Decision Tree**, **SVM**; counted reps at **95%+**.

Skills

- **Languages:** Python, JavaScript, TypeScript, C, C++, C#, Java, Scala, SQL, NoSQL, Go, Bash / Shell, Haskell, MATLAB, R
- **Frameworks & Libraries:** Next.js, Angular, Node.js, .NET, Django, Spring Boot, FastAPI, TensorFlow, PyTorch, OpenCV, Pandas
- **Tools & Infrastructure:** Git, GitHub, Linux, Docker, K8s, AWS, GitHub Actions, CI/CD, REST APIs, WebSockets, Postman